Assignment 1: Use Cases

Selected feature: Show the right ads to the right user at the right time

Course: CS3203 Fall 2026 | **Use case ID:** UC-04

Student name / ID: _____

Status: Draft to review and fill in before submitting.

Use case name

Show useful sponsored products while a shopper searches or browses.

Subject area

Online shopping and AI advertising.

Business event

A shopper searches for a product or opens a product category. This starts the process of choosing ads for that page.

Actors and system boundary

Actor / stakeholder	What they do
Shopper (primary actor)	Searches or browses products. The shopper does not need an account.
Outside services (supporting actors)	The ad service provides active ads and their budgets. The product service provides product details and stock levels. The privacy service tells the system what shopper data it can use.
Store and advertiser (stakeholders)	The store wants useful results and ad income. The advertiser wants shoppers to see its products. They do not need to take part in each search.

These services count as actors only if they are outside the shopping system.

System: The part of the shopping website that finds products and chooses ads. Its AI is part of the system, not a separate actor.

Use case overview

The system shows sponsored products that match what the shopper is looking for. Sponsored means the advertiser paid to promote the product. The system uses the current search or page and, when allowed, the shopper's past activity. It clearly labels ads and still shows normal product results if no useful ad is available.

Scope and assumptions

This is a proposed design for an example online store, not a description of how Constructor works behind the scenes. It covers a search or category visit, showing results, and an optional click on an ad. Setting up ads, paying, checking out, and training the AI are separate tasks. The privacy rules, ad limits, and time limits below are suggested choices for this design.

Preconditions and basic flow

Preconditions

1. The shopper can use the search box or category pages. Signing in is optional.
2. The system has product information and can show normal search results.
3. The store has rules for matching ads, labeling them, placing them, and using shopper data.
4. The system can check privacy settings. If it cannot confirm permission to use past activity, it uses only the current search or page. The process can start even if no ads are active.

Use case description: basic flow

1. **Shopper:** Searches for a product or opens a category. The shopper may add filters or choose a sort order.
2. **System:** Reads the search or category and checks what shopper data it is allowed to use.
3. **System:** Finds normal product results and possible sponsored products from active ads.
4. **System:** Removes ads that do not match the filters, are out of stock, have ended, have used up their budgets, or break store rules.
5. **System:** Uses AI to check how well each remaining ad matches the search or page. It uses past activity only when allowed. It removes ads that are not a good enough match.
6. **System:** Chooses the best ads using the store's goals and rules. An advertiser paying more does not let an unrelated ad pass the matching rules.
7. **System:** Checks again that the ads can still run. It shows the chosen products with a clear "Sponsored" label alongside normal results.
8. **Shopper:** Looks at the results and may click a sponsored product.
9. **System:** Opens the product page if the shopper clicks. When allowed, it records that an ad was shown or clicked. It does not share the shopper's personal browsing history with advertisers.

Termination outcomes and conditions: how the use case ends

Outcome	When it happens
T1 - Ads are shown	At least one ad matches the search and follows the rules. The results page loads. The shopper does not have to click an ad.
T2 - Only normal results are shown	No ad passes the checks, or the ad service fails while normal search still works. The shopper can continue shopping.
T3 - Results cannot be shown	The system cannot load the product results. It asks the shopper to try again.
T4 - Shopper leaves or changes the search	The shopper leaves or starts a new search before results appear. Ads that were never shown are not counted as views.

Alternative flows, inputs, and outputs

A1. Past activity cannot be used (step 2)

The shopper has not allowed the use of past activity, or that data is unavailable. The system uses only the current search or page and continues at step 3. If no ad matches, it shows normal results only (T2). A guest does not have to create an account.

A2. No suitable ad is found (steps 4-5)

All ads fail the matching, stock, budget, or store-rule checks. The system shows normal results only (T2). It does not show an unrelated ad just to fill a space.

A3. An ad or product changes before display (step 7)

A product goes out of stock, an ad ends, or its budget runs out. The system removes that ad and tries another suitable one if there is time. If none is available, it shows normal results only (T2).

A4. The ad service or AI does not respond (steps 3-6)

The service fails or takes too long. The system shows normal results without ads (T2). If normal search also fails, it asks the shopper to try again (T3).

A5. The search is invalid or finds nothing (step 1)

If the search cannot be used, the system asks the shopper to change it. If the search is valid but finds no products, it says no results were found and offers links to keep browsing. It does not show unrelated ads or count an ad view.

A6. The shopper changes the search or leaves (steps 7-8)

A new search or filter change starts a new check for suitable ads. The system stops old results from replacing the new results. If the shopper leaves before results appear, the use case ends with T4.

A7. A view or click cannot be recorded (step 9)

The shopper can still see products and open product pages. The system tries to record the event again when allowed. It gives each event its own ID so the same click is not counted twice. It records a click only if the shopper actually clicks.

Input summary

From the shopper: Search words, category, filters, sort order, product clicks, and privacy choices.
From other services: Product details, stock levels, ad dates, ad budgets, and store rules. Past shopping activity is used only when allowed and is not required.

Output summary

Normal results with zero or more products labeled "Sponsored"; a product page after a click; a no-results or try-again message when needed; allowed records of ad views and clicks; and an internal record of why an ad was chosen or skipped. Advertisers receive combined results, such as total views and clicks, rather than personal browsing histories.

Related use cases and responsibilities

Use case associations: related tasks

Search or browse products: Starts this use case.
Manage ads: Sets up ads before shoppers see them.
Manage privacy settings: Controls what shopper data can be used.
View product details: Happens after a product click.
Report ad results: Uses allowed view and click records. These are related tasks; this document does not define formal diagram links between them.

Traceability to: sources and requirements

Assignment: Assignment 1: Use Cases, Chapter Three, and Constructor use case 4.
Template: The IBM use-case template provided with the assignment.
Suggested requirements: R1 ads match the search and filters (steps 4-6); R2 ads are clearly labeled (step 7); R3 privacy choices are followed (steps 2 and 9, A1); R4 normal results still work if ads fail (A2-A4); R5 views and clicks are counted correctly (step 9, A7).

Usability index

Estimated score: 4.3 out of 5. This is the average of three suggested scores: shopper satisfaction = 4, importance = 5, and how often it is used = 4. The calculation is (4 + 5 + 4) / 3 = 4.33. These are estimates, not results from a survey. Shopper feedback, discussions with the store, and actual use would be needed to check them.

Use case notes: professional and ethical responsibilities

Good decisions: Ads should match what shoppers want and show products they can buy. The store should set clear rules for ad matching, the number of ads, and how long the system can wait.
Honesty and privacy: Label paid ads clearly. Collect only needed data and follow privacy choices. Do not give advertisers personal browsing histories or target shoppers using sensitive personal traits or guesses about them.
Fairness and access: Check whether different shoppers receive useful results. Make sure sponsorship labels are easy to read, including with tools such as screen readers.
Responsibility: Keep records of ad decisions with access limited to staff who need them. Review complaints and poor matches. Let store staff turn off ads that cause problems.

Simple checks

A matching, in-stock product should appear as a labeled ad. An unrelated ad should be rejected even if the advertiser pays more. Privacy choices should stop the use of past activity. Ads with no budget should be skipped. An ad-service failure should still allow normal results. Retrying a click record should not count it twice. The store should check shopper satisfaction as well as ad income.

References

[1] Constructor, "AI in Ecommerce: A Comprehensive Guide + 10 Use Cases," use case 4. constructor.com/blog/25-use-cases-for-ai-in-ecommerce/ (accessed October 5, 2026).
[2] IBM, "Use-case template," assignment PDF, updated March 14, 2024. IBM template reference.

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Add your name and student ID. Add what your group actually discussed. Check the course's Essential Standard Formatting Specifications and replace StudentID in the filename with your own ID. This draft does not claim that group discussion or peer review has been completed.

Group discussion notes: _____